

Keating: A Metaharness for Agency-Preserving AI Instruction

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August 08, 2026

AI tutors can scale explanation, but scaling explanation is not the same as scaling learning. A fluent answer may still weaken the learner’s own reconstruction of a concept. Keating is designed around that distinction: a **teaching metaharness** that coordinates live interaction, persistent learner evidence, inspectable artifacts, and evaluation-gated policy improvement. We analyze two deliberately separated evidence layers. An archival set of 22 model-generated sessions, deterministically curated to 16 topic x learner pairs, yields a normalized overall score of 0.61 (95% bootstrap interval 0.515-0.705) and reveals substantial topic heterogeneity and student-role contamination. A deterministic 14-topic benchmark shows that the frozen Keating 3.3.0 policy improves the default policy by 3.982 points on average (2.5th-97.5th percentiles: 3.039-4.985) across 200/200 seeds. In 30 isolated derivative-focused MAP-Elites reruns, selected policies reevaluated against the default on the same seed and fixed objective weights improve in 11, tie in four, and regress in 15 (mean delta -0.014). This exposes a mismatch between co-evolved search objectives and comparable scalar evaluation. These results support a systems-and-methods contribution, not a causal claim about human learning. The policy, traces, curated data, protocol, analysis outputs, and paper build are versioned for audit; a preregistered human trial remains the necessary next step.

Introduction

One-to-one tutoring remains one of the strongest known educational interventions (Bloom, 1984), and intelligent tutoring systems improve learning outcomes on average across domains (Ma et al., 2014). Recent work includes both a randomized trial of a structured generative-AI tutor (Kestin et al., 2025) and a deployed generative teaching assistant studied through use and learner perception (Thesen & Park, 2025). But tutoring quality cannot be judged only by whether the learner receives an explanation. Learning requires active reconstruction, self-explanation, retrieval, and transfer, whereas AI systems make it unusually easy for the learner to accept a polished account instead of generating one (Chi et al., 1989); (Rittle-Johnson & Loehr, 2017); (Roediger & Karpicke, 2006). This is exactly the terrain where cognitive offloading becomes a serious educational risk (Risko & Gilbert, 2016); (Burnett & Richmond, 2026).

Keating is built for that failure mode. The central idea is simple: if conventional AI tutors are mostly **chatbots that teach**, Keating is a **metaharness that governs teaching**. A chatbot is primarily an interaction policy at the level of the next response. Keating adds a control level above that response policy. It coordinates plans, maps, visual and interactive artifacts, verification checks, quizzes, flashcards, persistent learner state, prompt evolution, benchmark traces, and policy search across terminal, browser, and live multimodal teaching surfaces. The live exchange is therefore only one component of a larger pedagogical system. The controller also extends across time: it records lesson cadence, quiz and review evidence, and topic revisit urgency, using a simple retention-decay heuristic to propose spaced review. The educational motivation for spacing is well established (Cepeda et al., 2006), but Keating’s particular scheduling heuristic is not evaluated in this paper.

That distinction matters both scientifically and technically. For readers from learning science, the key claim is that the system is organized around diagnosis, retrieval, self-explanation, and transfer rather than answer delivery. For readers from machine learning and systems, the object being optimized is not merely a prompt or transcript but an **instructional environment** with explicit state, artifacts, objective functions, mutation operators, and acceptance boundaries. Recent systems work has begun to optimize harnesses and editable meta-level agents outside education (Lee et al., 2026); (Zhang et al., 2026). Keating applies that outer-loop framing to teaching while keeping provider-backed interaction, deterministic analysis, and real-learner evidence as distinct epistemic layers.

We make three contributions:

1. We formalize Keating as a teaching metaharness with explicit interaction, evidence, artifact, governance, and improvement layers.
2. We provide a mathematical and operational account of the deterministic fallback harness and the MAP-Elites/PROSPER policy-search path used in the study.
3. We present a versioned analysis stack that separates archival model-trace evaluation, internal synthetic stress testing, and the still-unmeasured question of human learning efficacy.

Keating as a Teaching Metaharness

Why a metaharness is different from a chatbot

A tutoring chatbot takes a learner message and emits a response. Its behavior is largely determined by a model, a prompt, and conversational context. A teaching metaharness also decides what instructional act should happen next, which artifacts and evidence should exist around that act, what counts as sufficient evidence for improvement, and how a future policy may be revised when the current one fails.

Keating is metaharnessed along five axes:

1. **Interaction layer.** Provider-backed text, dictation, full-duplex voice, camera, and screen-sharing sessions deliver the teaching exchange through web and terminal surfaces.
2. **Evidence layer.** Learner profiles, session history, feedback, quizzes, question checks, flashcard reviews, goals, and revisit priorities persist beyond a single response.
3. **Artifact layer.** The system produces lesson plans, concept maps, animations, verification records, study plans, decks, benchmark reports, prompt-evolution snapshots, and policy traces.
4. **Governance layer.** Teaching behavior is parameterized by explicit policy controls, and learner-facing evolution is blocked until at least five real outcome signals exist.
5. **Improvement layer.** MAP-Elites explores diverse policies, PROSPER-style comparison selects across multiple objectives, prompt evolution writes reviewable snapshots, and auto-improvement can roll back a regressing candidate.

In practical terms, Keating does not ask only “what should the tutor say next?” It also asks:

- What diagnostic and learner state should exist before explanation begins?
- What scaffold should require reconstruction rather than agreement?
- What evidence distinguishes a plausible response from retained or transferable understanding?
- When a learner returns after time away, which material should be retrieved before new material is introduced?
- Which changes may be accepted, rejected, or rolled back after evaluation?

Those are metaharness questions, not only chatbot questions.

Operational architecture and evidence boundary

The implementation has four operational layers. First, the Pi/OpenTUI runtime and React web agent run the live, provider-backed exchange. These paths may stream model text, structured tools, images, animations, audio, and video; they are intentionally non-deterministic and provider-dependent. Second, a shared tool and storage layer turns interactions into durable learner evidence and portable artifacts. Third, the local pedagogy core generates inspectable plans, maps, policies, traces, and deterministic fallback scores. Fourth, the improvement layer consumes that evidence through MAP-Elites, counterfactual evaluation when learner data exist, PROSPER-style preference, prompt evolution, and rollback-aware auto-improvement.

Several current product capabilities sit beside this core loop: browser-local model execution, remote sandbox routing, course workspaces, real-time collaboration, Anki import/export, and peer-to-peer course projection. They matter to delivery, privacy, and portability, but the present experiments do not evaluate their educational effect. Likewise, a successful build or deterministic benchmark is not evidence that a provider exchange, live audio path, collaborative room, or human-learning outcome works in deployment.

The production evidence boundary is stricter than the paper’s internal stress-test path. When `LearnerState` is supplied, the benchmark uses recorded feedback, quiz results, and inferred learner-turn signals; policy evolution refuses to run below five outcome records. Synthetic learners remain available only when no learner state is supplied, principally for tests and optimizer experiments. The study invokes that fallback deliberately and labels all resulting gains as **within-harness**.

The nearest systems analogues are frameworks that optimize harness code or editable self-improvement procedures outside education (Lee et al., 2026); (Zhang et al., 2026). Keating differs in the object being optimized: a pedagogical environment whose state, artifacts, objectives, and mutation rules are organized around diagnosis, retrieval, reconstruction, and transfer.

Natural entry points for different readers

Readers from education can view Keating as an attempt to operationalize a mastery loop: diagnose, teach, probe, repair, retrieve, and transfer. Readers from ML can view it as a structured controller over provider-backed instructional policies. Readers from systems can view it as an orchestration layer that moves some intelligence from hidden model behavior into inspectable state, artifacts, and objective functions.

The rest of the paper keeps those perspectives aligned. The results section reports what the archived and synthetic evidence can support. The methods section formalizes the benchmark and optimization protocol. The limitations section marks the larger live system as unevaluated wherever direct evidence is absent.

Results

The evidence stack

The paper uses two evidence layers.

1. **Archival model-trace evaluation.** We analyze model-generated teaching traces stored under `test/traces/`. These are not human-participant sessions.
2. **Internal deterministic benchmark.** We compare policies with the synthetic fallback implemented in `src/core/benchmark-real.ts` and orchestrated by `src/core/benchmark.ts`.

The layers answer different questions. The archival layer asks which concrete successes and failures appear in recorded model-to-model teaching sessions. The synthetic layer asks how policies and the current search procedure behave inside an explicit score model. Neither layer estimates a causal effect on human learning.

The repository contains 22 raw trace files, including repeated runs for some topic x learner pairs. We imposed a deterministic curation rule: retain the latest trace by timestamp for each pair. This yielded 16 retained sessions, matching the versioned snapshot `test/final_dataset.json`. The timestamp rule fixes the set by temporal provenance rather than selecting whichever repeated run looked best. One retained derivative trace for `Qwen-2.5-1.5B` contained `mastery=8`, `engagement=7`, and `clarity=8` while the rest of the archive used a 0-1 scale; we normalized that record to 0.8, 0.7, and 0.8 and recorded the correction in the generated bundle.

Component	Value	Interpretation
Raw archived traces	22	All preserved model-generated teaching transcripts before curation
Retained topic x learner pairs	16	Archival evaluation set used in this paper
Excluded duplicate earlier runs	6	Older runs for the same topic x learner pair
Score corrections	1 record	Single 10x encoding error normalized before aggregation
Synthetic topics	14	Internal benchmark tasks implemented in code
Synthetic learners per topic	3	Deterministic pseudo-learners sampled per topic and seed
Software snapshot	Keating 3.3.0	Version targeted by the frozen policy and analysis protocol

Table 1: Evidence layers, implementation snapshot, and curation rules.

Archival performance is heterogeneous across topics and learners

After normalization, the mean archival overall score, defined as the unweighted mean of mastery, engagement, and clarity, was 0.61 with a 95% bootstrap interval of 0.515-0.705. Performance varied sharply by topic. **Special Relativity** was strongest at 0.75 (0.596-0.883), followed by **Derivative** at 0.654 (0.454-0.767), **Social Contract Theory** at 0.613 (0.500-0.763), and **Stoicism** at 0.425 (0.283-0.558).

This pattern is substantively useful but not causal. The physics and calculus traces are structurally friendly to prediction, worked examples, and misconception repair, whereas the Stoicism traces demand introspective application. In this small archive, high instructional clarity does not reliably translate into demonstrated learner uptake on that introspective task.

Topic	n	Overall	Mastery	Interpretation
Special Relativity	4	0.750 (0.596-0.883)	0.695 (0.570-0.815)	Strong transfer from intuitive thought experiment to formal structure
Derivative	4	0.654 (0.454-0.767)	0.600 (0.263-0.788)	Conceptual calculus teaching is strong but not uniformly clean
Social Contract Theory	4	0.613 (0.500-0.763)	0.537 (0.462-0.650)	Mixed engagement and mixed transfer
Stoicism	4	0.425 (0.283-0.558)	0.287 (0.175-0.463)	Explanation often exceeds demonstrated learner uptake

Table 2: Archival model-trace evaluation by topic. Values are means with 95% bootstrap intervals.

Learner-model heterogeneity was also substantial. **Qwen-2.5-1.5B** scored highest overall at 0.779 (0.675-0.867), whereas **Llama-3.2-1B** scored lowest at 0.458 (0.367-0.550). We do not interpret these as broad claims about model families. The set is too small and the roles too artificial. The narrower result is that the teaching protocol was not uniformly robust across the four simulated learner profiles in the archive.

Student-role contamination is a central failure mode

The traces reveal a failure mode for agency-preserving instruction: student-role contamination. In some sessions, student turns begin to speak like a teacher or assistant rather than reconstructing the concept as a learner. Using a versioned heuristic over student turns, 5 of the 16 curated sessions showed at least one contamination marker.

These sessions performed worse descriptively. Sessions without contamination had mean mastery 0.575 and mean overall score 0.642, compared with 0.430 and 0.540 for contaminated sessions. The sample is small and the heuristic is not a validated classifier. Nonetheless, it demonstrates the kind of failure a metaharness can make visible: fluent text can coexist with weak evidence that the learner owns the idea.

The frozen policy is robust inside the current score model

The evaluated policy vector, **me-candidate-33**, is frozen in `docs/study/evaluated-policy.json`; the analysis does not read mutable `.keating` state. We compared that vector with the repository’s 3.3.0 default policy under the same default objective weights. This is a robustness comparison inside the current harness, not a held-out generalization result: the frozen candidate originated in full-suite MAP-Elites work.

Across 200 seeds, the frozen policy improved the default by 3.982 points on average (2.5th-97.5th percentiles: 3.039-4.985), winning on 200/200 aggregate suite comparisons. Its mean score was 59.039, compared with 55.057 for the default. Mean per-topic gains ranged from 3.142 points on **Legal Precedent** to 4.315 on **Cognitive Bias**; each topic improved on 196-200 of the 200 seeds.

The optimizer stability result is substantially more qualified. Thirty independent derivative-focused MAP-Elites runs used 24 candidates each and a fresh grid per run. Search candidates co-evolved policy coordinates and objective weights and were sampled on candidate-specific seeds, so their raw search scores are not directly comparable with the baseline. We therefore reevaluated each PROSPER-selected

policy and the default policy on the same run seed with `DEFAULT_WEIGHTS`. Under that standardized comparison, 11 selections improved, four selected the unchanged default, and 15 regressed. The mean delta was -0.014 , the median -0.164 , and the observed range -8.805 to $+7.142$. This protocol finds diverse candidates, but it does not show reliable scalar improvement under a fixed objective.

Synthetic analysis	Result	n	Interpretation
Frozen policy vs. default	+3.982 points (3.039-4.985)	200 seeds	Aggregate suite delta is positive on every sampled seed
Per-topic robustness	+3.142 to +4.315 mean	14 topics	Each topic is positive on 196-200 seeds
Independent derivative MAP-Elites reruns	11 wins, 4 ties, 15 regressions; mean -0.014	30 runs	Standardized reevaluation shows no reliable scalar gain

Table 3: Deterministic synthetic results for the frozen Keating 3.3.0 protocol.

Ablations show what the score model currently rewards

The frozen policy differs from the default in all nine controls. One-at-a-time swaps show that the largest gains come from lowering challenge rate (+2.227 points), maximizing retrieval practice (+1.820), and increasing interdisciplinary bias (+1.182). Increasing diagram bias to the frozen policy’s maximum produces the largest negative isolated effect (-1.583), while reflection bias contributes -0.376 in isolation.

These are properties of the score model and the tested policy coordinates, not estimates of pedagogical effect. The result suggests that the current synthetic surface is especially sensitive to overload control and retrieval. It also shows why joint search can be hard to interpret: a parameter that hurts in isolation may interact with other coordinates or objective weights inside a selected policy. An optimizer that can see this surface may exploit those sensitivities without producing better human teaching.

Parameter swapped into default	Mean synthetic delta	Interpretation
challengeRate	+2.227	Lower challenge reduces modeled overload most strongly
retrievalPractice	+1.820	The harness rewards enforced recall
interdisciplinaryBias	+1.182	Transfer-oriented prompting helps
analogyDensity	+0.657	Lower analogy density is weakly positive with a wide empirical range
exerciseCount	+0.343	Two rather than three exercises helps modestly
socraticRatio	−0.118	This coordinate is nearly neutral in isolation
formalism	−0.189	Additional formalism is slightly negative in isolation
reflectionBias	−0.376	Reflection emphasis is not rewarded in isolation
diagramBias	−1.583	Maximum visual emphasis is actively penalized in isolation

Table 4: One-at-a-time ablations diagnose the synthetic objective rather than human learning.

Discussion

The evidence supports three main claims.

First, Keating is usefully understood as a **teaching metaharness**. Its systems contribution is not simply a conversational interface, but an explicit control architecture around the exchange: persistent learner evidence, inspectable artifacts, policy parameters, evaluation paths, and reversible improvement procedures.

Second, the archival model traces yield meaningful failure analysis. High clarity can coexist with weak demonstrated mastery, introspective tasks were harder than formal technical tasks in this small archive, and student-role contamination provides a concrete example of fluent output without learner ownership.

Third, the frozen policy is robust inside the deterministic score model, but the optimizer result does not demonstrate reliable scalar improvement. The policy beats the default on every aggregate suite seed sampled. By contrast, when independently selected derivative-focused policies are reevaluated against the default on the same seed and fixed objective weights, only 11 improve, four tie, and 15 regress; the mean delta is -0.014 . That distinction separates evidence for one frozen policy from evidence about the search procedure that produced later candidates.

The standardized regressions expose a design issue in the current PROSPER-style selection path. Search co-evolves policy coordinates and objective weights, evaluates candidates on different seeds, and then chooses across scalar score, counterfactual robustness, mastery, transfer, low confusion, and evidence readiness. In a synthetic-only run, counterfactual and evidence-readiness signals are absent or uninformative, and a winner under its search conditions can underperform under a common objective and seed. Production reduces this risk by requiring at least five real learner outcomes before evolution

and by rolling back sufficiently large post-evolution regressions in the transactional auto-improvement loop. Those controls are implemented safeguards; they are not evidence that reward hacking or harmful policy drift has been eliminated.

The paper does **not** claim that Keating has been proven superior for human learners. The archive contains model-generated learner roles, not human participants; its labels are small, unblinded, and lack scorer provenance or inter-rater reliability. The synthetic benchmark is implemented and reproducible, but it remains a model of pedagogy rather than pedagogy itself. Nor do these experiments validate newer product capabilities such as live multimodal teaching, browser-local models, courses, collaboration, or spaced-review scheduling.

The next decisive study is a preregistered randomized comparison between Keating and at least one strong AI tutor baseline, with human participants, blinded rubric scoring, delayed retention tests, and explicit transfer tasks. In parallel, the benchmark should gain hidden holdout topics and coefficients, adversarial tests for degenerate strategies, and external transcript review. The goal is not merely to make policy search score better, but to make the link between the score and independent learner reconstruction harder to fake.

Methods

System overview

This revision targets Keating 3.3.0. The system has two related but distinct execution graphs. CLI, Pi, OpenTUI, and MCP surfaces converge on the Node project coordinator. The web app runs a browser-compatible Pi agent and a separately maintained browser port of the core, then routes workspace operations to browser-local, host, remote, or cloud adapters according to declared capabilities. The present paper evaluates neither interface graph directly; it focuses on the shared teaching-policy model, archived traces, and deterministic internal harness.

Within the Node path, lesson plans, maps, learner-state transforms, and fallback benchmark scores are local and inspectable. Some core operations remain model-assisted: animation authoring, factual verification, optional LLM judging, prompt optimization, and Ax/GEPA paths can call a provider and must surface a fallback or pending state. The paper’s synthetic results set `KEATING_LLM_BENCHMARK` off and use the algebraic fallback only.

A teaching policy contains nine scalar controls:

- analogy density
- Socratic ratio
- formalism
- retrieval practice
- exercise count
- diagram bias
- reflection bias
- interdisciplinary bias
- challenge rate

These controls do not encode a single answer. They define a region of instructional behavior. The metaharness evaluates that region against topic structure and learner evidence, then uses the result to compare or revise a policy.

LearnerState records sessions, topics, feedback, quiz results, goals, and review evidence. The engagement module converts last-seen time, session count, and mastery estimates into a revisit timeline using an exponential decay heuristic with configurable half-life, due threshold, minimum interval, and urgency tiers. This is motivated by evidence for distributed practice (Cepeda et al., 2006), but its calibration is not estimated from the present archive. More importantly, the production benchmark treats learner evidence and synthetic simulation differently: when learner state is supplied it scores recorded outcomes only, and policy evolution refuses to run until at least five feedback, quiz, or inferred learner-turn signals exist. The synthetic branch used below is an explicit internal fallback, not a substitute silently blended into a learner’s record.

Mathematical formulation of the harness

For readers from educational measurement, this section defines the latent teaching signals. For readers from ML systems, it specifies the benchmark objective. For readers from applied mathematics, it gives the explicit map from policy and learner parameters to session score.

Let a topic be represented by T , a policy by P , and a learner profile by L . The policy vector is

$$P = (a, s, f, r, e, d, b, i, c)$$

where a is analogy density, s Socratic ratio, f formalism, r retrieval practice, e exercise count, d diagram bias, b reflection bias, i interdisciplinary bias, and c challenge rate.

The learner vector is

$$L = (k, u, n, q, v, p, t, x)$$

where k is prior knowledge, u abstraction comfort, n analogy need, q dialogue preference, v diagram affinity, p persistence, t transfer desire, and x anxiety.

For a topic with formalism level φ_T and visualizability indicator ν_T , Keating computes the following fit terms. All intermediate quantities are clipped to the interval $[0, 1]$ after evaluation.

$$F_i = 1 - |a - n|$$

$$F_r = 1 - |f - \frac{\varphi_T + u}{2}|$$

$$F_d = 1 - |s - q|$$

$$F_g = 1 - |d - (\nu_T v + (1 - \nu_T)\omega_\nu)|$$

$$F_p = 1 - |\frac{e}{e_{\max}} - (1 - k + \omega_x x)|$$

$$F_b = 1 - |b - t|$$

Here **omega_nu** is the diagram fallback used for weakly visual topics, **omega_x** is the anxiety-to-practice coupling, and **e_max** is the exercise-count normalization constant induced by the policy domain. The model also computes an overload term parameterized by the bundle **Theta_0**:

$$O = \lambda_0 + \lambda_f f + \lambda_e \frac{e}{e_{\max}} + \lambda_c c - \lambda_p p + \lambda_x x - \lambda_k k$$

These intermediate quantities are then transformed into the five synthetic learning outcomes by the parameter bundles **Theta_M**, **Theta_R**, **Theta_E**, **Theta_T**, and **Theta_C**:

$$M = \mu_M + \alpha_i F_i + \alpha_r F_r + \alpha_d F_d + \alpha_g F_g + \alpha_p F_p + \alpha_o(1 - O)$$

$$R = M(\rho_0 + \rho_r r)$$

$$E = \mu_E + \beta_i F_i + \beta_d F_d + \beta_g F_g + \beta_b F_b + \beta_o(1 - O)$$

$$T_r = M(\tau_0 + \tau_i i + \tau_t t)$$

$$C = \mu_C + \gamma_o O + \gamma_f |f - u| + \gamma_c |c - p|$$

where M is mastery gain, R retention, E engagement, T_r transfer, and C confusion.

Finally, the session score is a weighted composition with bundle `Theta_S`:

$$S = \sigma_M M + \sigma_R R + \sigma_E E + \sigma_T T_r - \sigma_C C$$

Topic-level benchmark scores are the mean of S over the learner population for that topic, multiplied by a reporting scale parameter. Suite-level benchmark score is the mean over topics.

The harness is therefore defined by its structure plus its calibration:

$$\Theta = \{\omega_\nu, \omega_x, e_{\max}, \Theta_O, \Theta_M, \Theta_R, \Theta_E, \Theta_T, \Theta_C, \Theta_S\}$$

where each `Theta_*` denotes a small family of scalar parameters. The concrete coefficients are implemented in `src/core/benchmark-real.ts`; `src/core/benchmark.ts` selects between real-outcome scoring, deterministic simulation, and an explicitly enabled LLM judge. The equations present the fallback as a parameterized metaharness even though the 3.3.0 implementation fixes several values, including three synthetic learners per topic, a topic-seed stride of 97, and a reporting scale of 100. The formulation is interpretable rather than psychologically complete: its purpose is to make the path from policy coordinates to a score inspectable.

Harness pseudocode

```
BUILD-LEARNER-POPULATION(seed, count)
1 initialize PRNG with seed
2 learners <- empty list
3 for i <- 0 to count - 1
4   learner.id <- "learner-" + seed + "-" + i
5   learner.priorKnowledge <- RANDOM()
6   learner.abstractionComfort <- RANDOM()
7   learner.analogyNeed <- RANDOM()
8   learner.dialoguePreference <- RANDOM()
9   learner.diagramAffinity <- RANDOM()
10  learner.persistence <- RANDOM()
11  learner.transferDesire <- RANDOM()
12  learner.anxiety <- RANDOM()
13  append learner to learners
14 return learners
```

This procedure samples the synthetic learner population for one topic. Before Keating can evaluate a teaching policy internally, it needs a distribution of pseudo-learners with varying prior knowledge, abstraction comfort, persistence, and anxiety. The study uses the implementation's fixed count of three per topic. That is an engineering choice, not a claim that three simulated profiles approximate a human population.

```

SIMULATE-TEACHING(policy, topic, learner, theta)
1 intuitionFit <- 1 - |policy.analogyDensity - learner.analogyNeed|
2 rigorTarget <- CLIP((topic.formalism + learner.abstractionComfort) / 2)
3 rigorFit <- 1 - |policy.formalism - rigorTarget|
4 dialogueFit <- 1 - |policy.socraticRatio - learner.dialoguePreference|
5 if topic.visualizable
6   diagramTarget <- learner.diagramAffinity
7 else diagramTarget <- theta.visualFallback
8 diagramFit <- 1 - |policy.diagramBias - diagramTarget|
9 practiceNeed <- CLIP(1 - learner.priorKnowledge
10   + theta.practiceAnxietyWeight * learner.anxiety)
11 practiceFit <- 1 - |policy.exerciseCount / theta.exerciseNormalization
12   - practiceNeed|
13 reflectionFit <- 1 - |policy.reflectionBias - learner.transferDesire|
14 overload <- CLIP(theta.overloadBias
15   + theta.overloadFormalism * policy.formalism
16   + theta.overloadExercises
17   * policy.exerciseCount / theta.exerciseNormalization
18   + theta.overloadChallenge * policy.challengeRate
19   + theta.overloadPersistence * learner.persistence
20   + theta.overloadAnxiety * learner.anxiety
21   - theta.overloadKnowledge * learner.priorKnowledge)
22 masteryGain <- CLIP(theta.masteryBias
23   + theta.masteryIntuition * intuitionFit
24   + theta.masteryRigor * rigorFit
25   + theta.masteryDialogue * dialogueFit
26   + theta.masteryDiagram * diagramFit
27   + theta.masteryPractice * practiceFit
28   + theta.masteryHeadroom * (1 - overload))
29 retention <- CLIP(masteryGain
30   * (theta.retentionBase
31     + theta.retentionRetrieval
32     * policy.retrievalPractice))
33 engagement <- CLIP(theta.engagementBias
34   + theta.engagementIntuition * intuitionFit
35   + theta.engagementDialogue * dialogueFit
36   + theta.engagementDiagram * diagramFit
37   + theta.engagementReflection * reflectionFit
38   + theta.engagementHeadroom * (1 - overload))
39 transfer <- CLIP(masteryGain
40   * (theta.transferBase
41     + theta.transferInterdisciplinary
42     * policy.interdisciplinaryBias
43     + theta.transferDesire
44     * learner.transferDesire))
45 confusion <- CLIP(theta.confusionBias
46   + theta.confusionOverload * overload
47   + theta.confusionFormalismGap
48   * |policy.formalism - learner.abstractionComfort|
49   + theta.confusionChallengeGap
50   * |policy.challengeRate - learner.persistence|)
51 score <- CLIP(theta.scoreMastery * masteryGain
52   + theta.scoreRetention * retention
53   + theta.scoreEngagement * engagement
54   + theta.scoreTransfer * transfer
55   - theta.scoreConfusion * confusion)
56 return (masteryGain, retention, engagement, transfer, confusion, score)

```

This is the core scoring routine. It converts one policy-topic-learner triple into interpretable intermediate quantities and then into a final score. The important structural fact is that Keating does not score a policy directly. It first scores alignments: analogy pacing, rigor matching, dialogue matching, visual fit, practice load, reflection match, and overload. These are then composed into the five outcome variables

used by the benchmark. In that sense, the harness is factorized: it makes the path from policy parameters to session score inspectable. Passing the parameter bundle `theta` explicitly makes the generality of the metaharness visible. A reviewer can change the calibration without changing the algorithmic structure.

```
SUMMARIZE-TOPIC(topic, simulations, traceLimit, reportScale)
1 ranked <- simulations sorted in decreasing order by score
2 summary.meanScore <- reportScale
3   * MEAN(score for each simulation in simulations)
4 summary.meanMasteryGain <- MEAN(masteryGain for each simulation in simulations)
5 summary.meanRetention <- MEAN(retention for each simulation in simulations)
6 summary.meanEngagement <- MEAN(engagement for each simulation in simulations)
7 summary.meanTransfer <- MEAN(transfer for each simulation in simulations)
8 summary.meanConfusion <- MEAN(confusion for each simulation in simulations)
9 summary.topLearners <- first traceLimit entries of ranked
10 summary.strugglingLearners <- last traceLimit entries of ranked, reversed
11 summary.dominantStrength <- strongest average alignment signal
12 summary.dominantWeakness <- weakest average alignment signal
13 return summary
```

This procedure aggregates a set of learner-level simulations into a topic-level result. The benchmark preserves both population averages and diagnostic tails. The mean score describes average behavior under the score model; the strongest and weakest pseudo-learners expose which fit terms dominate. The pseudocode leaves the reporting scale explicit, while the current implementation fixes it at 100.

```
RUN-BENCHMARK-SUITE(policy, topics, seed, weights, learnerState?, config)
1 outcomes <- learnerState ? EXTRACT-REAL-OUTCOMES(learnerState) : empty
2 if learnerState exists and outcomes is not empty and topics is not focused
3   topics <- unique topics represented in outcomes
4 topicBenchmarks <- empty list
5 for j <- 0 to length(topics) - 1
6   topic <- topics[j]
7   if learnerState exists
8     topicOutcomes <- outcomes matching topic
9     simulations <- topicOutcomes is empty
10      ? empty
11      : [SCORE-REAL-OUTCOMES(topicOutcomes, policy, topic, weights)]
12   else
13     learners <- BUILD-LEARNER-POPULATION(seed + 97 * j, 3)
14     simulations <- map learners through SIMULATE-TEACHING(policy, topic, learner, weights)
15     summary <- SUMMARIZE-TOPIC(topic, simulations, config.traceLimit, 100)
16     append summary to topicBenchmarks
17 weakestTopic <- topic with minimum meanScore in topicBenchmarks
18 overallScore <- MEAN(meanScore for each topic summary in topicBenchmarks)
19 dataSource <- learnerState exists ? real-evidence status : synthetic
20 return (overallScore, weakestTopic, topicBenchmarks, dataSource)
```

This is the current benchmark boundary. A caller that supplies learner state does not receive a synthetic blend: for an unfocused benchmark, only topics represented in real outcomes are scored; an explicitly focused topic without outcomes produces an empty summary. A caller that omits learner state enters the deterministic fallback used by tests and the present study. The returned trace labels the data source so a user-facing report can distinguish sufficient evidence, sparse evidence, no evidence, and synthetic execution.

The user-facing product coordinator applies an additional gate before calling the search routine: if learner state is supplied, at least five real outcomes must be available. `mapElitesEvolve` itself is a lower-

level primitive and does not enforce that threshold, so direct research and Ax callers remain responsible for their own evidence boundary.

```
MAP-ELITES-EVOLVE(basePolicy, focusTopic, iterations, seed, grid, learnerState?)
1 baseline <- RUN-BENCHMARK-SUITE(basePolicy, focusTopic, seed,
2   DEFAULT_WEIGHTS, learnerState)
3 place baseline in grid(formalism, socraticRatio)
4 candidates <- [baseline]
5 initialize PRNG with seed
6 for iteration <- 1 to iterations
7   if iteration is in the initial random quarter
8     policy, weights <- RANDOM-POLICY-AND-WEIGHTS(PRNG)
9   else
10    parent <- uniformly sampled filled grid cell
11    policy, weights <- MUTATE(parent.policy, parent.weights, PRNG)
12    benchmark <- RUN-BENCHMARK-SUITE(policy,
13      focusTopic,
14      seed + 11 * iteration,
15      weights,
16      learnerState)
17    if learnerState exists
18      counterfactual <- BENCHMARK-PERTURBED-OUTCOMES(policy, weights, learnerState)
19    replace the matching grid cell if empty or benchmark score is higher
20    append (policy, benchmark, counterfactual) to candidates
21 winner <- PROSPER-PAIRWISE-WINNER(candidates,
22   objectives = [score, counterfactual robustness, mastery,
23     transfer, low confusion, evidence readiness])
24 persist grid and decision ledger
25 return (baseline, winner, grid, candidates)
```

This is the current default improvement loop. It does not train model weights. It explores policy and objective-weight coordinates, retains the best score within each behavioral grid cell, and then applies a PROSPER-style pairwise preference across all candidates. Diversity and final selection are therefore separate: MAP-Elites maintains a repertoire, while PROSPER chooses one policy. Unlike the older hill-climbing implementation, the final preference does not impose a monotone scalar-score or weakest-topic gate. The standardized regressions reported in the results are evidence of that distinction, not noise to be hidden.

External archival evaluation

We analyzed the 22 JSON trace files stored in `test/traces/`. Because multiple traces existed for some topic x learner pairs, we retained the chronologically latest trace for each pair, yielding 16 sessions spanning four topics (Derivative, Special Relativity, Stoicism, and Social Contract Theory) and four learner models (Llama-3.2-1B, LFM-2.5-1.2B, Qwen-2.5-1.5B, and Cloud-MiniMax-M2.5). The retained set exactly matched `test/final_dataset.json`. This timestamp-based rule was chosen to make archival evaluation deterministic and auditable: later reruns replace earlier reruns by a visible provenance field rather than by manual judgment.

Each retained trace already contained three scalar labels: mastery, engagement, and clarity. We treated these as archived outcome labels. One record encoded scores on a 0-10 scale rather than the 0-1 scale used elsewhere, so we normalized that record by dividing by 10 and recorded the correction in `docs/generated/study-analysis.json`.

The external overall score was defined as the unweighted mean of mastery, engagement, and clarity. Because the dataset is small, we report bootstrap intervals rather than formal null-hypothesis tests.

We also computed exploratory trace features, including empty turns, word counts, teacher redirection cues, and student-role contamination markers.

Synthetic benchmark and robustness analyses

The internal benchmark uses `src/core/benchmark.ts` with `KEATING_LLM_BENCHMARK` disabled, which routes simulation to the algebraic model in `src/core/benchmark-real.ts`. Unless a focus topic is specified, the suite evaluates 14 topics. For each topic and random seed, it samples three pseudo-learners and computes topic means for mastery gain, retention, engagement, transfer, confusion, and their weighted score.

The evaluated vector is the exact JSON snapshot at `docs/study/evaluated-policy.json`, frozen from the Keating 3.3.0 `me-candidate-33` state. The analysis compares it with `DEFAULT_POLICY` under `DEFAULT_WEIGHTS` over seeds 1-200. Because that candidate originated in full-suite work, per-topic comparisons are robustness checks, not held-out tests. To probe mechanism, we swapped each frozen-policy coordinate individually into the default and reevaluated the same 200 seeds.

To probe search stability, we ran `mapElitesEvolve` 30 times from `DEFAULT_POLICY`, focused on **Derivative**, with 24 candidates and seeds 1-30. Every run received a distinct temporary grid path, preventing persisted elites from leaking between reruns. Candidate search scores use co-evolved objective weights and candidate-specific seeds, so we retain them only as provenance. For the reported delta, the selected policy and `DEFAULT_POLICY` are reevaluated on the same run seed with `DEFAULT_WEIGHTS`. Production normally uses 48 candidates and a persistent 10 x 10 grid; the smaller isolated study protocol is a bounded diagnostic, not a reconstruction of the frozen policy’s original search history.

We did **not** run a dedicated reward-hacking study in which the optimizer was challenged against hidden holdout metrics, adversarially perturbed harness coefficients, or external human judgments. That omission is important because any policy-search loop can in principle exploit regularities in the scoring function rather than improve the underlying pedagogical behavior the score was meant to represent.

Statistics and reporting

All derived numbers in the manuscript come from `bun scripts/study-analysis.mjs`, exposed as the `keating:study-analysis` devenv task. The script writes the versioned `docs/generated/study-analysis.json` and `docs/generated/study-analysis.md`; the JSON includes protocol metadata, every seed result, and every optimizer rerun. The marimo notebook at `analysis/study_analysis.py` provides an inspectable analysis surface. External descriptive intervals use 5,000 non-parametric bootstrap resamples with deterministic pseudo-random seeds. Synthetic robustness summaries are empirical means, percentiles, and observed counts across seeds or isolated reruns; they are not confidence intervals over human learners.

Limitations

The strongest limitation is population validity. No human participants were studied. The archival “learners” are language models assigned a student role, and the labels do not include scorer provenance, blinded review, or inter-rater agreement. The archive supports descriptive systems analysis only.

A second limitation is harness validity. The deterministic benchmark strongly rewards lower modeled overload and retrieval practice, while maximum diagram bias and higher reflection bias are negative in isolation. Those sensitivities are properties of the implemented equations. They are not validated measures of human mastery, retention, engagement, transfer, or confusion.

A third limitation is optimizer safety. In 30 isolated MAP-Elites reruns, standardized reevaluation of the PROSPER-selected policy regressed 15 times, tied four times, and improved 11 times, with a mean delta near zero. Co-evolved objective weights and candidate-specific seeds can make raw search scores look favorable without establishing improvement under a common comparison. Production adds a real-evidence threshold and transactional rollback, but this study does not evaluate those protections against adversarial or long-horizon policy drift.

A fourth limitation is reward hacking. The optimizer is evaluated against an inspectable score surface, and we did not test hidden metrics, perturbed coefficients, adversarial policies, or external human judgments. An optimizer may learn benchmark regularities more quickly than it learns pedagogy.

A fifth limitation is data volume. Sixteen retained traces across four topics and four learner models are enough to surface failure cases, but not enough for stable cross-domain or model-family claims. The timestamp curation rule prevents cherry-picking among duplicates; it does not make the resulting sample representative.

A sixth limitation is evaluation lag relative to the product. Keating 3.3.0 includes persistent quizzes and flashcards, revisit scheduling, live audio/video teaching, browser and remote execution modes, collaborative courses, and portable study artifacts. None of those subsystems receives a direct educational-outcome evaluation here. The engagement timeline is motivated by distributed-practice evidence (Cepeda et al., 2006), but its exponential decay model, thresholds, and priority lanes remain engineering heuristics.

Near-term mitigations are concrete: maintain hidden holdout topics and coefficient settings; require candidate policies to pass scalar, weakest-topic, and multi-objective gates; evaluate against multiple harness variants; add adversarial strategies such as superficial retrieval spam and low-challenge gaming; version every study input; and require periodic blinded human audit before describing a policy update as better teaching.

Code and Reproducibility Availability

Keating is available under the Mozilla Public License 2.0 at <https://github.com/Diogenesoftoronto/keating>. The system snapshot targeted by this revision is version 3.3.0. Study logic is contained in `scripts/study-analysis.mjs`, `analysis/study_analysis.py`, `src/core/benchmark.ts`, `src/core/benchmark-real.ts`, `src/core/map-elites.ts`, and `src/core/policy-judgement.ts`.

The evaluated policy is frozen at `docs/study/evaluated-policy.json`; analysis no longer reads mutable `.keating` state. Run `devenv tasks run keating:study-analysis` to regenerate `docs/generated/study-analysis.json` and `docs/generated/study-analysis.md`. Run `devenv tasks run keating:paper` to regenerate the analysis and compile the published PDF at `web/public/keating-metaharness.pdf`. `keating:paper-check` performs a compile-only check to a temporary path.

Data Availability

The 22 raw model-generated traces are versioned under `test/traces/`, and the deterministic 16-record curation snapshot is `test/final_dataset.json`. The archive contains no human-participant data. `docs/generated/study-analysis.json` records the curation manifest, the one score normalization, all external summaries, the frozen protocol, every synthetic seed comparison, ablations, and all 30 isolated optimizer reruns.

The paper’s quantitative claims can therefore be regenerated from a fresh checkout without provider credentials. Live provider calls, deployed browser behavior, audio/video transport, collaborative-course operation, and human-learning outcomes are outside that reproducibility claim.

Bibliography

- Bloom, B. S. (1984). The 2 Sigma Problem: The Search for Methods of Group Instruction as Effective as One-to-One Tutoring. *Educational Researcher*, 13(6), 4–16. <https://doi.org/10.3102/0013189X013006004>
- Burnett, L. K., & Richmond, L. L. (2026). Meta-analytic Investigations of the Effect of Cognitive Offloading on Memory-based Task Performance and Interindividual Variability. *Memory & Cognition*, 54(1), 144–168. <https://doi.org/10.3758/s13421-025-01743-8>
- Cepeda, N. J., Pashler, H., Vul, E., Wixted, J. T., & Rohrer, D. (2006). Distributed Practice in Verbal Recall Tasks: A Review and Quantitative Synthesis. *Psychological Bulletin*, 132(3), 354–380. <https://doi.org/10.1037/0033-2909.132.3.354>
- Chi, M. T. H., Bassok, M., Lewis, M. W., Reimann, P., & Glaser, R. (1989). Self-Explanations: How Students Study and Use Examples in Learning to Solve Problems. *Cognitive Science*, 13(2), 145–182. [https://doi.org/10.1016/0364-0213\(89\)90002-5](https://doi.org/10.1016/0364-0213(89)90002-5)
- Kestin, G., Miller, K., Klales, A., Milbourne, T., & Ponti, G. (2025). AI Tutoring Outperforms In-Class Active Learning: an RCT Introducing a Novel Research-Based Design in an Authentic Educational Setting. *Scientific Reports*, 15, 17458. <https://doi.org/10.1038/s41598-025-97652-6>
- Lee, Y., Nair, R., Zhang, Q., Lee, K., Khattab, O., & Finn, C. (2026). Meta-Harness: End-to-End Optimization of Model Harnesses. *Arxiv Preprint Arxiv:2603.28052*. <https://doi.org/10.48550/arXiv.2603.28052>
- Ma, W., Adesope, O. O., Nesbit, J. C., & Liu, Q. (2014). Intelligent Tutoring Systems and Learning Outcomes: A Meta-Analysis. *Journal of Educational Psychology*, 106(4), 901–918. <https://doi.org/10.1037/a0037123>
- Risko, E. F., & Gilbert, S. J. (2016). Cognitive Offloading. *Trends in Cognitive Sciences*, 20(9), 676–688. <https://doi.org/10.1016/j.tics.2016.07.002>
- Rittle-Johnson, B., & Loehr, A. M. (2017). Eliciting Explanations: Constraints on When Self-Explanation Aids Learning. *Psychonomic Bulletin & Review*, 24(5), 1501–1510. <https://doi.org/10.3758/s13423-016-1079-5>
- Roediger, H. L., & Karpicke, J. D. (2006). Test-Enhanced Learning: Taking Memory Tests Improves Long-Term Retention. *Psychological Science*, 17(3), 249–255. <https://doi.org/10.1111/j.1467-9280.2006.01693.x>
- Thesen, T., & Park, S. H. (2025). A Generative AI Teaching Assistant for Personalized Learning in Medical Education. *Npj Digital Medicine*, 8, 627. <https://doi.org/10.1038/s41746-025-02022-1>
- Zhang, J., Zhao, B., Yang, W., Foerster, J., Clune, J., Jiang, M., Devlin, S., & Shavrina, T. (2026). Hyperagents. *Arxiv Preprint Arxiv:2603.19461*. <https://doi.org/10.48550/arXiv.2603.19461>